bandwidth extending to the frequency of oscillation only.

Method 3: Use amplifiers in "Switch mode" to switch the output between two set voltage levels together with some form of time delay to control the time at which the amplifiers switch-on or off, thus controlling the period of the signal produced.

Methods 1 and 2 are used extensively in sine wave oscillators.

Method 3 is used in square wave generators (multivibrator).

7.4.4. Constant Amplitude

As shown in Figure 7.4, oscillators must have an amplifier, a positive feedback loop and some method of controlling the frequency of oscillation. In RF sine wave oscillator, the frequency may be controlled by an LC tuned circuit, but as well as controlling the frequency of oscillation, there must be some means, such as negative feedback for stabilizing the amplitude of signal produced.

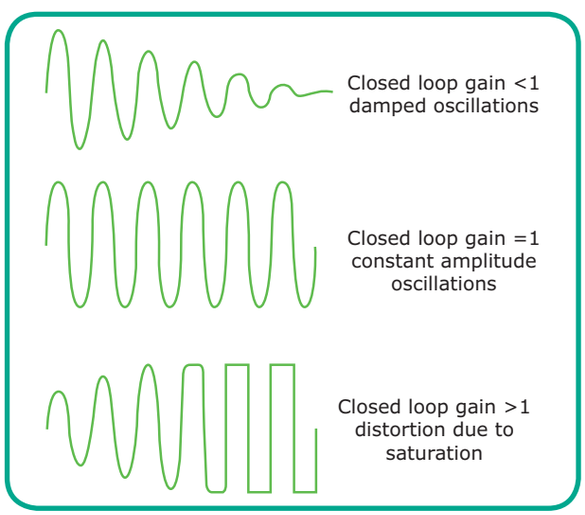


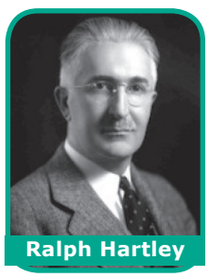
Figure 7.5 Constant Amplitude

Without this stabilization, the oscillations would either die away and stop (damped oscillation) or rapidly increased in amplitude until the amplifier produces severe distortions due to the transistors within the amplifier becoming "saturated" as shown in Figure 7.5. To produce constant amplitude output the gain of the amplifier is automatically controlled during oscillation.

7.5. TYPES OF OSCILLATOR

There are several types of transistor oscillators commonly used in electronic circuits. Let us study the few important types. 1. Hartley Oscillator, 2. RC Phase Shift Oscillator, 3. Crystal Oscillator

7.5.1. Hartley Oscillator


Ralph Hartley

This type of LC oscillator is used to produce RF waves in the range of 30 kHz - 30 MHz The frequency of oscillation is decided by its tank circuit, which has a capacitor (C) connected in parallel with inductors (L_1 and L_2).

History: The Hartley oscillator circuit was invented in 1915, by Ralph Hartley. The distinguishing feature of the circuit is that the tuned circuit consists of a single capacitor in parallel with two inductors in series, and the feedback signal needed for oscillation is taken from the center connection of two inductors.

Construction: Figure 7.6 shows the Hartley oscillator. The circuit consists of CE configuration, in which Resistor R_1 and R_2 form the voltage divider bias network. Next, the capacitor C_{C2} and C_{C1} are input and output decoupling capacitor, while the emitter capacitor C_E is the bypass capacitor used to bypass the amplified AC signals.

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